

Linux Practical Test. Time:1 Hours

Chapters: Basic commands, Hard link soft link, User management, Permission management.

1. Create a directory named projectdata inside /opt.
Inside it create 3 files: dev.txt, test.txt, and prod.txt.
2. Create a directory /securedata. Create a file secret.txt inside it. Only the owner should have full permissions. No permissions should be given to group or others.
3. Create 5 empty files named: server1.log to server5.log inside /var/log/testlogs using a single command.
4. Create a user named devopsuser with:
Home directory /training/devopsuser
Bash shell
User ID 2001
Set password to DevOps@123.
5. Create a group named projectteam.
Create 3 users:: rahul, aman, zoya

Add all users to the projectteam group as secondary members.
6. Create a user named backupadmin with:

Account expiry date: 2026-12-31
Password expiry after 15 days

7.Lock the user account rahul.Then unlock the same account and verify its status.

8.Create a shared directory /sharedproject with these conditions:

Owner group should be projectteam

All group members should be able to create/.
deletefiles

Other users should not access the directory.

9.Create a user named audituser.

Set these password policies:

Password expiry: 20 days

Warning before expiry: 5 days

10.Create a group named admins.Allow members of this group to run all commands using sudo without asking for a password.

11.Create a user named temporaryuser.

Set these restrictions:

User should not get interactive shell login

User should not have a home directory

12.Create a backup of:

/etc/passwd

/etc/shadow

/etc/group

Store all backups inside /root/userbackup with a .bak extension.

13.Create a directory /teamdata.

Create a file project.txt inside it.

Give:

Read and write permission to user rahul

Read-only permission to user aman
using ACL.

14.Create a file named finance.txt.

Set permissions so that:

Owner has full permissions

Group has read-only

Others have no permissions

Display permissions in numeric format.

15.Change the system umask temporarily so that:

Newly created files get permission 640

Newly created directories get permission 750

Create one file and one directory to verify.

16.Create a user named operator.

Configure permanent umask for this user so that:

New files get 600

New directories get 700

Verify after login.

17.Create a directory /publicdata.

Apply Sticky Bit permission so that:

Users can create files

Only file owners can delete their own files

Verify the permissions.

18.Create a file important.conf.

Give execute permission only to the owner using symbolic method.
Remove all permissions from others.

19.Create a directory /acltest.

Inside it create two files:

dev.txt

prod.txt

Using ACL:

Give user rahul full permission on dev.txt

Give user aman only read permission on prod.txt

20.Create a file named database.txt inside /opt/project.

Create:

One hard link named db_hard

One soft link named db_soft

21.Create a soft link named webconfig pointing to /etc/httpd/conf/
httpd.conf

22.Create the following users, groups, and group memberships.

A group named sysadmin. A user ryan who belongs to sysadmin as
a secondary group. A

user sarah who also belongs to sysadmin as a secondary group. A
user harry who does
not have access to an interactive shell on the system, and who is
not a member of

sysadmin. ryan, sarah and harry should all have their password as
atenorth.

23. Create a Collaborative Directory

Group ownership of /common/admin is to be set to sysadmin. The directory should be readable, writable, and accessible to members of sysadmin, but not to any other user. (It is understood that root has access to all files and directories on the system.)

Files created in /common/admin should automatically have group ownership set to the sysadmin group

24. User of Specific UID

Create a user james UID of this user should be 2112 and set password as: sestiver

25. Set Password Expiry

The password for all new users should expires after 20days.

26. Working with Strings

Copy all words with 'wired' from /usr/share/dict/. words to the file /tmp/data

27. Set Umask Permanently

Set default permissions of rahul user after creating a file be "r--r--r--" and after creating a directory to be "r-xr-xr-x

28. Rename the dev.txt file with new name should be

grass.txt the file already exists inside projectdata.

29. Apply the sticky bit to a directory

/etc/passwd to prevent users from deleting
others' files ?

30. Create a directory /secureproject with these conditions:

- Group name should be devteam
- User rahul and aman must be members of the group
- Owner should have full permissions
- Group members should have read/write/execute permissions
- Others should have no permissions
- Apply SGID on the directory
- Apply Sticky Bit on the directory
- Set default ACL so user aman always gets read/write permission on newly created files
- Set temporary umask so newly created files become 640